



# **“STUDENT ACADEMIC PERFORMANCE PREDICTION BY USING MACHINE LEARNING”**

**By**

**WAI YAN PHYO**

**Student No: 20048955**

**MASTER OF SCIENCE IN INFORMATION SYSTEM WITH  
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**Supervisor: DR. STEPHEN AZEEZ**

**Dublin Business School**

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## Dedication

This thesis is dedicated to my parents and family for their unconditional love, support, encouragement, and motivation throughout my academic career. They have inspired me to successfully complete this research by their sacrifices, patience, and belief in my abilities.

## Acknowledgement

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## Abstract

A prediction of student academic performance has emerged as one of the key areas of research in learning analytics and educational data mining. In this study, machine learning methods will be examined to predict student academic performance based on demographic data, academic, and behavioral data. The data set used in research is Student Performance Data retrieved from Kaggle which consists of data regarding students' education background, study habits, prior academic achievement, and social characteristics.

Decision Tree, Random Forest and Logistic Regression three machine learning algorithms were implemented and evaluated to predict student academic outcomes. Several pre-processing techniques were applied before the development of the model, such as data cleaning, label encoding, feature selection, feature reduction, and train-test splitting. The most important predictive variables were determined using the method of correlation analysis, Chi-square testing, and Random Forest feature importance.

The results of the study indicated that the academic performance variables that predicted final academic performance (G3) were the highest with previous academic performance variables (G1 and G2). Other factors that were also significant were absences, failures, and age. The Logistic Regression model was the best model in terms of predictive performance (Accuracy, Precision, Recall, F1-Score, and AUC-ROC) among the evaluated models. The results show that most of the predictive relationships in the data are linear.

The study highlights the predictive value of machine learning solutions in the education sector and shows how well machine learning methods can predict students' academic achievement. The results could inform early identification of academically at-risk students and to inform evidence-based educational decision making.

The growing amount of data available for the education sector is enabling the development of tools for analyzing student performance and learning patterns, a field known as learning analytics.

|                                                                         |    |
|-------------------------------------------------------------------------|----|
| CHAPTER 1 .....                                                         | 2  |
| 1.1 Background of the Study .....                                       | 3  |
| 1.2 Research Motivation .....                                           | 4  |
| 1.3 Problem Statement .....                                             | 5  |
| 1.4 Research Questions .....                                            | 5  |
| 1.5 Aim and Objectives .....                                            | 6  |
| 1.6 Limitation of the Study .....                                       | 6  |
| 1.7 Scope of the Study .....                                            | 7  |
| Chapter 2 – Literature Review .....                                     | 8  |
| 2.1 Introduction .....                                                  | 8  |
| 2.2 Theoretical Framework in student performance .....                  | 9  |
| 2.3 Machine Learning Approaches in Student Performance Prediction ..... | 9  |
| 2.4 Early Warning System (EWS) Models .....                             | 11 |
| 2.5 Prior Studies on Performance Prediction .....                       | 12 |
| 2.5.1 Comparative Summary of Prior Studies .....                        | 13 |
| 2.6 Research Gaps .....                                                 | 13 |
| 2.7 Summary .....                                                       | 14 |
| Chapter (3) .....                                                       | 15 |
| 3.1 Introduction .....                                                  | 15 |
| 3.2 Research Design and Philosophy .....                                | 15 |
| 3.3 Dataset Description .....                                           | 16 |
| 3.4 Feature Selection .....                                             | 16 |
| 3.5 Model Selection and Justification .....                             | 19 |
| 3.6 Model Training and Hyperparameter Tuning .....                      | 20 |
| 3.7 Evaluation Metrics .....                                            | 21 |
| 3.8 Summary .....                                                       | 22 |
| Chapter 4 – Results & Analysis .....                                    | 23 |
| 4.1 Introduction .....                                                  | 23 |
| 4.2 Preprocessing and Data Summary Outcomes .....                       | 23 |
| 4.3 Model Performance .....                                             | 24 |
| 4.3.1 Decision Tree .....                                               | 24 |

|                                                                    |    |
|--------------------------------------------------------------------|----|
| 4.3.2 Random Forest.....                                           | 26 |
| 4.3.3 Logistic Regression .....                                    | 28 |
| 4.4 Cross-Model Analysis .....                                     | 30 |
| Comparative Model Performance Table.....                           | 30 |
| 4.5 Feature Importance Analysis.....                               | 32 |
| 4.6 Summary .....                                                  | 34 |
| Chapter 5 – Conclusions, Recommendations & Future Directions ..... | 34 |
| 5.1 Conclusion.....                                                | 34 |
| 5.2 Recommendation.....                                            | 36 |
| 5.3 Future Directions .....                                        | 37 |
| Reference.....                                                     | 38 |

# CHAPTER 1

## 1.1 Background of the Study

Academic achievement is one of the most accepted measures of academic success in higher education institutions. Student academic performance is constantly measured in universities and colleges due to the significant relationship it has with the retention of the student, the rates of graduation, the reputation of the institution and the quality of education it provides. Academic probation, delayed graduation, or student dropouts can have negative consequences for both the student and the educational institution and may stem from poor academic performance. As a result, research into the factors affecting student academic performance has become a major area of research in modern education systems.

Educational institutions are increasingly making and keeping vast quantities of data related to students and the learning environment. The field of educational technology and digital learning environments is developing rapidly. These datasets include demographics, attendance, last academic performance, behavioral trends, learning styles, and social influences. Typically, such data were analyzed using statistical techniques. Traditional analytical methods, however, might not be suitable to handle complex and non-linear relationships between variables related to student achievement (Huang and Fang, 2013).

Over the past few years, with the advent of Educational Data Mining (EDM) and the new field of machine learning techniques, the analysis of educational datasets and the prediction of learning outcomes has become very powerful. In its simplest terms, Educational Data Mining involves using computational and analytical tools to analyze educational data for meaningful patterns to assist in educational decision-making (Baker and Inventado, 2014). Given historical data, machine learning algorithms can learn, and create predictive models that can uncover patterns linked to student success or failure.

The application of machine learning algorithms in predicting student academic performance has been widely accepted. Data mining techniques were shown to be useful to predict students' achievement with high accuracy by Kabakchieva (2013), where demographic and academic variables were used. Moreover, other researchers such as Asif et al. (2017) stated that Educational Data Mining can help in educational institutions to identify students at risk of poor education performance in early stages. Marbouti, Diefes-Dux, and Madhavan (2016) also highlighted the need for predictive models to identify students at risk academically and to support early interventions. Different machine learning techniques have been used for educational prediction tasks: Decision Tree, Random Forest, Logistic Regression, Support

Vector Machine, and Neural Networks. These models can be used to interpret student learning trends and to enhance academic monitoring and decision making by the educational institution.

Of these, Decision Tree models are known for being interpretable, Random Forest models are known to have better predictive capability due to ensemble learning and Logistic Regression offers efficient and interpretable classification performance (Kumar and Singh, 2019). Hence, in this study, the three predictive models: Decision Tree, Random Forest, and Logistic Regression are compared with the Student Performance dataset from Kaggle to determine the effectiveness of the models.

## 1.2 Research Motivation

This study is motivated by both academic and practical concerns in relation to students' academic success and the effectiveness of institutions. Academic performance prediction is an important research area in education data mining, machine learning, and learning analytics, as many educational institutions are adopting data-driven methods to enhance the academic performance of their students.

The need for conducting this study comes from the growing difficulty that schools experience when they try to identify the students that are likely to perform poorly in their academic performance before the release of their exam results. The conventional methods used in such cases normally depend on teacher evaluation, post-examination evaluation, or observation. This often leads to failure in providing timely help to those students that need further assistance.

To tackle this challenge, a machine learning approach can be employed to process the historical data of the education fields and create predictive models that help in early identification of students at risk. Early prediction systems can help educational institutions put in place timely interventions, like academic counseling, tutoring, mentoring, and customized academic support. Early alert systems can make a significant contribution to the success of institutional efforts in supporting academically at-risk students, Jayaprakash et al. (2014) write.

Educational Data Mining has become an indispensable tool in modern education, which has been another motivation for this study. Many institutions are still not leveraging these educational data for predictive and analytical uses, while others are still collecting huge amounts of educational data. By leveraging machine learning algorithms on educational data, institutions can gain insights into learning patterns, improve educational planning, and optimize educational decision-making processes.

Moreover, machine learning algorithms have been shown to be effective for high predictive accuracy in educational prediction tasks (Costa et al., 2017). But still comparative studies



comparing different machine learning models with publicly available educational datasets are required. Hence, the motivation of this study is to compare the effectiveness of different algorithms like Decision Tree, Random Forest, and Logistic regression in the prediction of student academic performance.

### 1.3 Problem Statement

Higher Education Institutions continue to face accurate identification of students at risk of poor performance before final examination results are released remains a challenge for higher education institutions. The traditional academic monitoring systems are typically based on manual evaluation and simple statistical analysis, which may not capture the complexity of the interactions between the demographic, behavioral, and academic variables.

Without effective predictive systems, intervention strategies may be delayed, students may fail to graduate, they may leave the school, and institutional performance may decrease. Students at risk may not be identified in a timely fashion, as educational institutions may not detect warning signs at an early stage. As a result, poor academic learning results can take place before intervention measures are implemented.

Although several studies have applied machine learning techniques in educational environments, there remains a need to evaluate and compare multiple machine learning algorithms using structured educational datasets. Furthermore, some studies focus primarily on prediction of accuracy without sufficiently analyzing influential features affecting student academic performance.

This study aims to solve these problems by creating and testing machine learning models that predict academic performance of the undergraduate students based on the Kaggle Student Performance Dataset. The study will evaluate the performance of the algorithms Decision Tree, Random Forest, and Logistic Regression to see which one is best suited for predicting students at risk of poor academic performance.

### 1.4 Research Questions

The purpose of this study is to respond to the following research questions:

1. Which variables in the Kaggle Student Performance Dataset significantly influence undergraduate student academic performance?
2. How effective are machine learning techniques in predicting undergraduate student academic performance?
3. Within Decision Tree, Random Forest, and Logistic Regression, which machine learning model provides the highest predictive performance?
4. How does feature selection influence the performance of machine learning models in student performance prediction?
5. Can machine learning models effectively identify students who are at risk of poor academic performance before final examination results are released?

## 1.5 Aim and Objectives

This study aims to develop and test machine learning models to predict performance of undergraduate student academics by utilizing the Kaggle Student Performance Dataset.

The following objectives are suggested to realize the research goal:

- To analyze the dataset and find out the important variables affecting the students' academic performance.
- To clean and preprocess the dataset, including handling categorical variables and applying feature reduction.
- Implement feature selection techniques to obtain the most relevant features to predict.
- To evaluate the impact of feature selection on model performance and identify the most influential predictors of student academic outcomes
- To assess and compare the performance of the selected machine learning models on accuracy, precision, recall, f1 score, AUC\_ROC, and confusion matrix.
- To determine which machine learning model is best suited to predict students at risk of poor academic performance.

## 1.6 Limitation of the Study

There are some limitations that should be considered when interpreting the results of this study.

The first thing is that the study is based on the publicly available dataset from Kaggle, and it has been taken from the UCI Machine Learning Repository. As a result, the data sets may not be representative of all undergraduate student populations in various institutions, countries, and educational systems.

Second, the data are limited in number of variables and do not include psychological, emotional, motivational, or mental health factors that can also be of influence on a student's academic

performance. In addition, there isn't classroom interaction data or real-time learning management system data in the dataset.

Thirdly, the analysis in this research only includes three machine learning algorithms: Decision Tree, Random Forest, and Logistic Regression. In this study, machine learning techniques such as Support Vector Machines, Gradient Boosting, XGBoost, and Deep Learning models other than the ones mentioned above are not considered.

The other constraint is that the research is only dependent on the analysis of secondary data, without any primary data collection process from actual students or institutions. This means that the quality and structure of the dataset may have a direct impact on the results of the research.

One important limitation of this study is the inclusion of G1 and G2 as predictive features. Since these variables are highly correlated with the final grade (G3), the predictive performance of the models may appear higher than in scenarios where only demographic and behavioral variables are available.

## 1.7 Scope of the Study

The main aim of this study is to determine the academic performance of the undergraduate students by predicting it using machine learning methods on the Kaggle Student Performance Dataset.

Within the scope of the study, preprocessing of data, feature selection, development of predictive model, model training, hyperparameter tuning, and performance evaluation are covered. In particular, the dataset will be cleaned and then the categorical variable will be encoded, relevant features will be selected, the dataset will be divided into training and testing sets, and predictive models will be developed with Decision Tree, Random Forest and Logistic Regression algorithms.

In order to measure the performance of the models, some performance measurement metrics are considered, which include accuracy, precision, recall, F1 score, AUC-ROC, and confusion matrix. Also, feature importance analysis will be used in determining the significant variables that have an impact on student performance.

However, the study does not include the introduction of a real institutional Early Warning System, longitudinal student tracking, real-time education monitoring, or advanced deep learning approaches. Research is limited to the context of structured educational data, in predictive modelling and comparative evaluation.

## Chapter 2 – Literature Review

### 2.1 Introduction

The prediction of student academic performance has become a significant research topic in the field of education, data science, and learning analytics. As more educational datasets become available, researchers are able to use machine learning techniques to identify patterns and predict student outcomes in a more efficient way.

The paper will be based on the **Student Performance Dataset available on Kaggle (Cortez & Silva, 2008)**, which consists of student-level data with detailed variables such as demographic, academic, and behavioral. The dataset includes features such as study time, number of past failures, parental education level, and social behavior indicators.

The prediction target in this study is the **final grade (G3)**, which is the general achievement of the students in the academic field. Additionally, intermediate grades (G1 and G2) are included as predictive features.

To model this problem, three machine learning algorithms are chosen, such as Decision Tree, Random Forest, and Logistic Regression. These models are selected because they can manage structured educational data and provide a balance between interpretability and prediction.

The main research problem is:

***Which machine learning model provides the most accurate prediction of student final academic performance (G3) using the Kaggle Student Performance Dataset?***

The following parts of this chapter are also organized to support the selection of dataset, variables, and machine learning models.

## 2.2 Theoretical Framework in student performance

Student academic performance is influenced by a combination of cognitive, behavioral, and environmental factors. These influences can be understood through established theoretical frameworks.

According to Bandura's Social Cognitive Theory (1986), self-efficacy is very important in learning. Students who have belief in their ability to be successful are more likely to engage in academic activities and achieve higher outcomes. Failure, studytime, G1/G2 scores are some of the variables that may be understood as signs of academic confidence and academic performance history in the context of this dataset. However, since self-efficacy itself is not directly measured, it limits the full theoretical application of the framework.

Vygotsky's Zone of Proximal Development (1978) mentions the importance of support in learning. Students perform better when guided appropriately. In this dataset, **parental education (Medu, Fedu)** and **family support variables (famsup)** can represent academic support structures. A limitation is that detailed classroom interaction data is not available.

Socio-economic theories further explain that external factors influence student performance. Socio-economic factors like family size, parental occupation, and access to the internet are variables that represent the socio-economic conditions which can influence learning outcomes (Cortez & Silva, 2008). There is, however, a problem of these factors, not capturing individual motivation or psychological differences.

## 2.3 Machine Learning Approaches in Student Performance Prediction

Machine learning methods are especially appropriate to analyze educational data because they can model complex relationships among multiple variables.

The machine learning models in this research are trained to make a prediction of G3 (final grade) using the following features:

- Study time (time spent studying)
- failures (past academic failures)

- parental education (Medu, Fedu)
- absences (attendance behavior)
- G1 and G2 (previous grades)

The reason why the Decision Tree model is chosen is because it is interpretable. It enables the easy visualization of the influence of such variables as studytime or failures on the ultimate performance. Nevertheless, it may overfit the data (Kabakchieva, 2013).

Random Forest is better than Decision Trees because it uses several trees which minimize overfitting and enhances accuracy. Studies indicate that Random Forest is effective in learning data (Romero & Ventura, 2020).

Logistic Regression is provided as a baseline model. In our case, G3 can be converted into categories (e.g., pass/fail) and thus, Logistic Regression should be used when performing classification. However, its assumption of linear relationships can restrict performance.

Even though high accuracy can be obtained by using models such as Support Vector Machines, Neural Networks and Gradient Boosting (Kumar & Singh, 2019), they are not chosen because

- Lower interpretability
- Higher computational complexity
- Less suitability for small/medium datasets

In such a way, the chosen models offer a balance between accuracy, interpretability, and practicality. Several previous studies have reported strong predictive performance using machine learning techniques for student performance prediction. Kabakchieva (2013) reported a highest classification accuracy of 66.59% using the J48 Decision Tree classifier. Similarly, Cortez and Silva (2008) reported that previous academic grades, particularly G1 and G2, significantly improved prediction accuracy in educational datasets.

Marbouti, Diefes-Dux and Madhavan (2016) demonstrated that Random Forest and Logistic Regression models could effectively identify academically at-risk students with high predictive reliability. More recently, Kumar and Singh (2019) observed that advanced machine learning techniques such as Support Vector Machines and Neural Networks may achieve higher predictive accuracy; however, these methods often involve greater computational complexity and lower interpretability. Compared to previous studies, the present research emphasizes not only predictive performance but also model interpretability and practical applicability within educational environments.

An important methodological consideration in this study is the inclusion of G1 and G2 as predictive features. Previous educational data mining research has consistently shown that prior

academic grades are among the strongest predictors of future academic performance (Cortez and Silva, 2008; Asif et al., 2017). However, the inclusion of G1 and G2 also introduces a known limitation because both variables are highly correlated with the target variable G3. As a result, the predictive models may achieve higher classification accuracy partly due to the availability of near-proxy variables for the final academic outcome. Therefore, the findings of this study should be interpreted as prediction based substantially on interim academic performance indicators in addition to demographic and behavioral factors. Nevertheless, G1 and G2 were retained because the study aims to evaluate practical predictive performance within realistic educational settings where previous academic records are commonly available and widely used for student monitoring and intervention.

## 2.4 Early Warning System (EWS) Models

Early Warning Systems (EWS) aim to identify students at risk of poor academic performance and enable early intervention.

In this dataset, risk indicators include:

- Low G1 and G2 scores
- High number of failures
- Low study time
- High absences

Predictive models can be used to determine at-risk students before definitive results are available by using these variables. This study does not implement a complete Early Warning System (EWS). Rather, the research focuses only on the predictive component, which identifies academically at-risk students using machine learning models before final examination results are released.

A complete Early Warning System generally requires continuous student monitoring, institutional intervention mechanisms, and structured support services to assist students who are identified as vulnerable to poor academic performance. In educational settings, such systems are designed not only to predict academic risk but also to provide timely academic assistance and intervention strategies.

The work of Asif et al. (2017) and Marbouti et al. (2016) shows that the achievement of students who are at risk of failing academically can be enhanced greatly by making some predictions about their performance at an early stage. But if no institutional mechanisms and support are provided, predictive models cannot be used as a complete EWS.

## 2.5 Prior Studies on Performance Prediction

There is a large amount of literature on applying data mining and machine learning methods to the prediction of student academic performance. Previous research by Cortez and Silva (2008) is especially important as it presented the student performance dataset with which this study works. Their study revealed that earlier grades (G1 and G2), time spent on the study, and the number of previous failures is good predictors of the final academic performance (G3). This research serves on a direct basis to the present research since it verifies the dataset, as well as the relevance of the variables.

Other studies have been conducted with the aim of comparing machine learning algorithms. Kabakchieva (2013) compared various methods of classification and discovered that Decision Trees are highly interpretable, although they might not be predictive. This brings out the trade-off between model transparency and performance, which is a significant factor in educational research.

Much more recent studies have focused on the power of ensemble techniques. In their robust review of educational data mining, Romero and Ventura (2020) found that one of the most reliable models is the Random Forest because it is strong and can minimize overfitting. On the same note, Sweeney, Lester, and Rangwala (2016) also revealed that ensemble techniques coupled with feature selection strategies greatly enhance the accuracy of predictions.

Moreover, Asif et al. (2017) used predictive analytics in higher education and demonstrated that predictive analytics can positively impact academic performance by early detecting at-risk students and intervening promptly. Their results indicate the significance of predictive modeling in decision-making in learning institutions.

This opinion is also supported by the research of Marbouti, Diefes-Dux, and Madhavan (2016), who prove that it is possible to predict final performance with the help of early indicators in the academic sphere. Their research revealed that it is possible to determine students who may fail by using variables gathered at the beginning of a course.

Moreover, Kumar and Singh (2019) evaluated various machine learning models, such as Support Vector Machines and Neural Networks, and discovered that advanced models tend to be more accurate, although more computationally resource-intensive and less interpretable. This supports the need to choose models that would be both performing and interpretable.

Another topical research by Huang and Fang (2013) emphasized the fact that previous academic achievement is one of the most significant predictors of success in the future. This result is especially applicable to this research, in which G1 and G2 are the predictors of G3.



All in all, these studies indicate that machine learning methods are useful predictors of student performance, but they also show that there are significant trade-offs in accuracy, interpretability, and practical applicability.

### 2.5.1 Comparative Summary of Prior Studies

| Study                              | Dataset                                              | Methods Used                 | Key Findings                                                                                                                                               | Limitations                                   |
|------------------------------------|------------------------------------------------------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| <b>Cortez &amp; Silva (2008)</b>   | Student dataset (Portugal)                           | Decision Tree, Regression    | G1, G2, studytime (strong predictors)                                                                                                                      | Limited generalization                        |
| <b>Kabakchieva (2013)</b>          | Educational dataset                                  | Multiple classifiers         | DT interpretable but less accurate                                                                                                                         | Lower predictive performance                  |
| <b>Romero &amp; Ventura (2020)</b> | Literature review of educational data mining studies | Ensemble learning methods    | Surveyed studies report that ensemble methods including Random Forest consistently achieve strong predictive performance in student performance prediction | No direct empirical experimentation conducted |
| <b>Asif et al. (2017)</b>          | Higher education data                                | Predictive analytics         | Early prediction improves outcomes                                                                                                                         | Requires institutional data                   |
| <b>Marbouti et al. (2016)</b>      | Course-level data                                    | Machine learning             | Early indicators predict final grades                                                                                                                      | Context-specific                              |
| <b>Sweeney et al. (2016)</b>       | Large dataset                                        | Ensemble + feature selection | Improved accuracy                                                                                                                                          | Computational cost                            |
| <b>Kumar &amp; Singh (2019)</b>    | Student dataset                                      | SVM, NN, DT                  | Accuracy vs interpretability trade-off                                                                                                                     | High complexity                               |
| <b>Huang &amp; Fang (2013)</b>     | Engineering education                                | Regression/ML                | Prior performance strongest predictor                                                                                                                      | Limited features                              |

## 2.6 Research Gaps

Although much has been done in terms of research on the prediction of student performance, there are several important gaps remaining.

Firstly, numerous studies use institution-specific or privately collected data, which limits reproducibility and comparability. There is a clear need for research by using publicly available datasets such as the one used in this study.

Second, despite the application of different machine learning algorithms, there is limited research that offers a direct comparison of Decision Tree, Random Forest, and Logistic Regression on the same dataset and evaluation parameters.

Thirdly, while predictive student performance in the undergraduate domain has been extensively explored in the field of educational data mining, there are not many research studies that have looked at publicly available and structured educational data and that have placed emphasis on both predictive accuracy and interpretability of the models. One aspect of interpretability is especially relevant in education, where teachers, administrators, and decision makers must be able to comprehend the factors that affect the prediction, and to be able to support meaningful academic interventions and decision-making processes.

Fourth, although early prediction has been widely studied, many models are not directly applied to undergraduate-level datasets or do not fully explore the predictive power of variables such as prior grades (G1 and G2).

Finally, there is limited integration between predictive modeling and practical applications such as early identification of at-risk students using simple and interpretable models.

This study addresses these gaps by applying multiple machine learning models to a publicly available dataset and focusing on both predictive performance and interpretability.

## 2.7 Summary

This chapter provided a review of the available literature on the topic of student performance prediction, such as theoretical frameworks, machine learning, early warning systems, and previous research.

The review points out that academic achievements of students are determined by a variety of factors and machine learning methods are useful in predicting academic performance. The models that have been shown to be useful in the case of previous research include Decision Tree, Random Forest, and Logistic Regression.

Nevertheless, the research gaps remain especially in the application of publicly available datasets, comparative model testing, and practical implementation in the identification of at-risk students.

Therefore, this study builds upon existing research by applying and comparing machine learning models to predict undergraduate student exam performance using the Kaggle Student Performance Dataset.

## Chapter (3)

### 3.1 Introduction

This chapter will introduce the methodology that will be used in this research in order to predict the academic performance of students using a machine learning technique. The research is based on the systematic research method, which involves data collection, data cleaning, preprocessing, feature selection, model development, hyperparameter tuning, and evaluation of performance. The data sample of this study includes student level data, such as demographic, academic, and behavioral data. The main goal is to predict the end student grade (G3) which is converted to a binary classification problem (pass or fail) to enable the use of classification algorithms.

To achieve my objective, three machine learning models are being used to implement such as Decision Tree, Random Forest, and Logistic Regression. All the models are trained and tested with the help of proper techniques to ensure reliability and accuracy. The methodology will be structured to facilitate reproducibility, robustness, and compliance with the common practices in educational data mining and machine learning.

### 3.2 Research Design and Philosophy

This study takes a quantitative research design because the study will use numerical data and statistical analysis to determine the relationship between different student-related variables and academic performance. The philosophy of the research is based on the positivist philosophy, which supposes that the knowledge is possible to be obtained due to objective measurement and empirical evidence. Applying to machine learning models makes it possible to identify the patterns in the dataset and predict the student outcomes based on the things that can be observed.

The dataset utilized in this paper has 395 student records and 33 attributes. These characteristics are academic indicators like past grades (G1 and G2) and behavioral factors (absences) and socio-demographic factors (parental education). The analysis is based on a predictive modeling

format in which the independent variables are used to predict the dependent variable (G3). The overall research design will help achieve data-driven and generalizable findings that can be applied to similar educational contexts.

### 3.3 Dataset Description

The data set used in this research work was collected from Kaggle, which had been taken from the UCI Machine Learning Repository (Cortez and Silva, 2008). The data consists of 395 student records and 33 variables, such as demographic, academic, family, and behavioral data. Variables include school, sex, age, address, famsize, Pstatus, Medu, Fedum, Mjob, Fjob, reason, guardian, traveltime, studytime, failures, schoolsup, famsup, paid, activities, nurse, higher, internet, romantic, famrel, freetime, goout, Dalc, Walc, health, absences, and previous academic grades (G1 and G2).

The target variable in this study is G3, which represents the final student grade. For classification purposes, G3 was converted into a binary variable representing pass and fail outcomes. Students with G3 values greater than or equal to 10 were categorized as “pass,” while students with grades below 10 were categorized as “fail.”

After preprocessing and feature selection, the final dataset consisted of 20 features used for model training and evaluation.

### 3.4 Feature Selection

The task of feature selection is essential to enhance model performance, considering the most relevant variables and minimizing noise in the data. The three feature selection methods used in this study are correlation analysis, Chi-squared testing, and Random Forest feature importance.

Correlation analysis was conducted to determine the linear relationships between numerical variables and target variables (G3). The findings inspect that G2 is positively correlated with G3 with a correlation coefficient of about 0.90. On the same note, G1 also demonstrates a robust positive correlation of about 0.80. These results indicate that the past academic success of students is a high predictor of their ultimate success. Conversely, there is a negative correlation between the variable failures and G3 such that students with more past failures tend to have lower final grades.

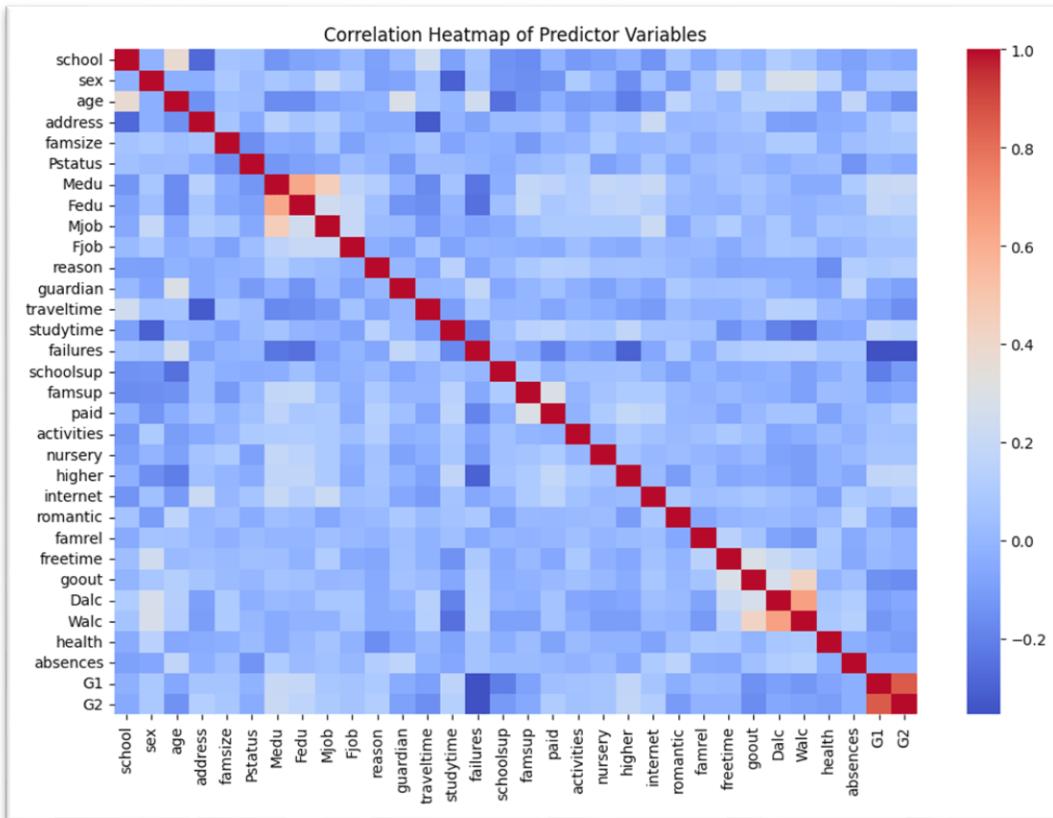
The Chi-squared test was used to further determine the importance of categorical and non-linear relationships. The results demonstrate that G2 has the highest Chi-square value (approximately 273.8), followed by G1 (approximately 175.9) and failures (approximately 74.4). Also, absences also exhibit a medium impact with a Chi-square value of about 37.6. These findings confirm that

academic and behavioral variables have significant contributions to the prediction of student performance.

The importance of each feature was then evaluated using the Random Forest feature importance to evaluate the contribution of each feature to the prediction model. The results show that G2 has the highest importance with an importance score of about 0.3629, then G1 (0.2065), absences (0.0480), age (0.0302), and failures (0.0299). These results reinforce the findings done through correlation analysis and Chi-square analysis.

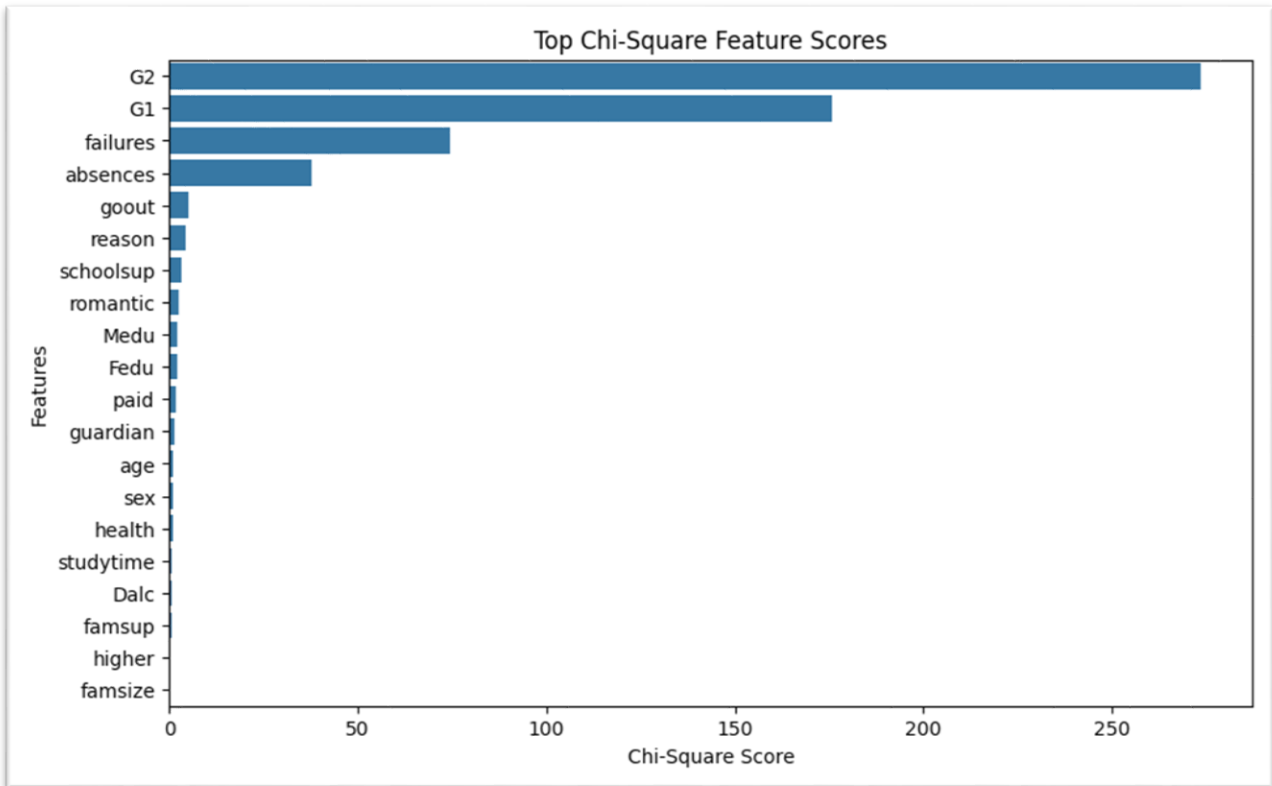
These findings led to the feature reduction by dropping the variables with a very low importance score (less than 0.01). The eliminated features entailed variables such as travel time, engagement of extracurricular activities, access to the internet, and other social features. This step made the models less complex, while maintaining predictive performance.

Although G1 and G2 significantly improve predictive performance, their inclusion introduces an important methodological consideration because both variables are highly correlated with the target variable G3. In particular, the correlation coefficient between G2 and G3 is approximately 0.90, indicating that a substantial proportion of the variance in final academic performance can already be explained by previous academic grades alone. Consequently, the high predictive accuracy achieved by the models is influenced partly by the availability of near-proxy variables for the target outcome. Therefore, the findings of this study should be interpreted as prediction based partly on interim academic performance indicators rather than solely demographic and behavioral factors. Nevertheless, G1 and G2 were retained in this study because previous academic performance is theoretically and practically recognized as one of the strongest predictors of future student achievement in educational data mining research (Cortez and Silva, 2008; Asif et al., 2017).



*Figure 3.1 Correlation Heatmap*

Figure 3.1 presents the correlation heatmap of the numerical variables used in this study. The heatmap visually demonstrates the strong positive relationships between G1 and G2, while variables such as failures exhibit negative correlations with final academic performance.



*Figure 3.2 Chi Square Feature Scores*

Figure 3.2 illustrates the Chi-square feature scores obtained for the most influential variables. The results indicate that G2 and G1 contribute to the highest statistical importance, followed by failures and absences.

### 3.5 Model Selection and Justification

This study selected three machine learning models on their appropriateness to classification problems and their capacity to utilize structured data.

The Decision Tree model was chosen because it is easy to interpret and can be used to represent decision rules in a hierarchical form. It is especially applicable to educational research, where it is of importance to understand the impact of variables. But Decision Trees are susceptible to overfitting more so when the depth of the tree is uncontrolled.

The Random Forest model was selected as the further development of the Decision Tree model. It is an ensemble learning technique that involves using several decision trees to enhance the predictive accuracy of the decision tree and minimize overfitting. Random Forest would be a good fit for this dataset since it can deal with complicated interactions between variables and is

resistant to noise. Nevertheless, the model is not as interpretable as a single decision tree would be.

The choice of a baseline model was made based on the simplicity of Logistic Regression, as well as its efficiency in binary classification problems. It simulates a student with the input features having a probability of passing or failing a student. Even though it assumes that there is a linear relationship between the independent variables and the log-odds of the dependent variable, it can be useful in providing a comparison to evaluate more complex models.

### 3.6 Model Training and Hyperparameter Tuning

An 80:20 split of the dataset was done to form a training and a testing set. The parameter of `random_state` (42) was fixed so that the results could be reproducible. The models were implemented in Python 3.12.13 and the scikit-learn library.

In order to have robust model evaluation, a 5-fold cross-validation was applied during training. The Decision Tree model achieved an average cross-validation F1-score of approximately 0.9186 while the Random Forest model achieved the highest cross-validation F1-score of approximately 0.9473. Logistic Regression produced an average cross-validation F1-score of approximately 0.9405. The relatively small differences between cross-validation performance and testing performance across the three models indicate acceptable generalization capability.

The grid SearchCV was used for hyperparameter tuning to determine the best configuration of each model. The best parameters determined to be used with the Decision Tree model were a maximum depth of 10, a minimum sample split of 10 and a minimum samples leaf of 4. In the case of the Random Forest model, the optimal parameters were 100 estimators, a maximum depth of 20, and the square root of features were used as splitting functions. In the case of Logistic Regression, the best regularization parameter (C) was discovered to be 10.

As it offers a tradeoff between precision and recall and is applicable to classification problems where both false positives and false negatives are of interest, the F1-score was taken as the main measure of hyperparameter tuning.

The hyperparameter tuning process illustrated that there was an acceptable degree of sensitivity for the models, especially tree-based models, with respect to how well they performed. The Decision Tree model showed that higher maximum depths led to smaller F1-scores and overfitting, and greater depths did not result in significant increases in the F1-scores.

The model was chosen to have a maximum depth of 10 to balance the generalization and accuracy of the model. Likewise, the Random Forest model demonstrated relatively stable



performance for a variety of parameter settings, with more estimators leading to greater stability and lower variance of the model. For Logistic Regression, higher values of the regularization parameter (C) slightly improved the performance of the classifiers by avoiding underfitting. In conclusion, hyperparameter tuning played a positive role in yielding better consistency in the models and improved predictive performance with all three machine learning algorithms.

### 3.7 Evaluation Metrics

The machine learning models were tested with various classification metrics such as Accuracy, Precision, Recall, F1-score, AUC-ROC, and Confusion Matrix. These evaluation metrics were chosen due to their ability to accurately measure the classification performance and effectively compare the performance of the classification models.

Accuracy is the ratio of the number of instances correctly classified by the model to the total number of predictions made by the model. It tests the success of the overall classification model.

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

Meaning:

TP = True Positives

TN = True Negatives

FP = False Positives

FN = False Negatives

Precision is the ratio of the number of positive instances that are correctly predicted to the number of instances that are predicted to be positive. When minimizing false positive predictions is a priority, precision is important.

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Recall (also known as sensitivity): It is the ratio of true positives to total positives. In the education sector, recall is crucial as it can be used to ensure that those at risk of failing academically are identified.

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

The F1-Score is a balance of Precision and Recall. This is particularly useful when there are both false positives and false negatives.

$$\text{F1-score} = (2 \times \text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

The Area Under the Receiver Operating Characteristic Curve (AUC-ROC) measures the performance of the model over a range of classification thresholds. The larger the AUC value, the better the classification ability and the higher the discriminative power.

The Confusion Matrix is a detailed report of classification results with a number of correct and incorrect classifications. It lets you examine in detail the subsets of True Positives, True Negatives, False Positives, and False Negatives created by each predictive model.

These evaluation metrics together give a holistic view about how well these three models, Decision Tree, Random Forest and Logistic Regression, predict student academic performance.

This dataset has moderate class imbalance which means that there are more students who are classified as pass than fail, and accuracy is not enough to assess the performance of the model in this study. Thus, other measures like Precision, Recall, and F1 score were needed to have a more balanced evaluation of the predictive performance. Moreover, the use of the F1-score and of the AUC-ROC allows for the assessment both with and without the introduction of a threshold (or thresholds) and therefore provides a more robust and dependable measure of the effectiveness of a classification.

These evaluation metrics give a comprehensive picture of the accuracy of the student's academic performance prediction models, Decision Tree, Random Forest, and Logistic Regression.

### 3.8 Summary

In this chapter, the methodology employed for predicting student academic performance using machine learning techniques was described. The research process included data cleaning, preprocessing, feature selection, feature reduction, model training, hyperparameter tuning, and evaluation metrics. Three classification models, namely Decision Tree, Random Forest, and Logistic Regression, were implemented and evaluated using multiple classification metrics.

The chapter also discussed the importance of previous academic performance, attendance behavior, and other demographic and behavioral variables in student performance prediction. Furthermore, the methodology for model evaluation, cross-validation, and hyperparameter optimization was presented to ensure reliable and reproducible results.

Overall, this chapter has presented the complete methodological framework adopted for the study. The following chapter presents and analyzes the experimental results obtained from the implemented machine learning models and discusses their implications within the context of educational data mining and student performance prediction.

## Chapter 4 – Results & Analysis

### 4.1 Introduction

The chapter includes the findings and discussion of the machine learning models that were created to predict the academic performance of students. The analysis is done on the performance of the chosen models: Decision Tree, Random Forest, and Logistic Regression according to their predictive capabilities. This chapter starts with a summary of the results of data preprocessing, and then the results of model performance are presented in detail, along with cross-model comparisons. Moreover, the analysis of feature importance is performed to determine the most significant factors that influence student performance. The results are viewed against the objectives of the research and theoretical frameworks discussed in Chapter 2.

### 4.2 Preprocessing and Data Summary Outcomes

The data that was utilized in this study was subjected to various preprocessing procedures, which included the following: data cleaning, encoding, feature selection, and feature reduction. The dataset had 395 instances and 34 features after encoding the categorical variables. In contrast with one-hot encoding which increases the number of features, label encoding was used in this study; hence, the number of features did not grow significantly as compared to the original number of variables.

After performing feature selection using Random Forest feature importance, low-importance features were removed based on a threshold criterion of importance less than 0.01. The deleted variables were nursery, activities, romantic, famsize, paid, internet, famsup, address, sex, higher, school and Pstatus. Following the reduction process, the final dataset consisted of 395 instances, and 20 features, as all unnecessary variables were removed during the reduction process and the most relevant variables were retained for model training.

The dataset was next divided into training and testing set through a stratified sampling method with random seed (`random_state = 42`). This made sure that the proportions of the target variable in classes remained the same in both subsets. The number of training sets was 316 instances, and the number of test sets is 79 instances; this represents a split ratio of 80:20.

The analysis of the class distribution validates the fact that the dataset is moderately imbalanced and consistent across splits. In the entire dataset, about 67.1% of students are in the class of pass and 32.9% are in the class of fail. The training set has a similar distribution with about 67.4% pass and 32.6% fail and the test set has approximately 65.8% pass and 34.2% fail. This shows that stratified splitting was an effective way of maintaining the original proportions of the classes.

It is important to note that the variables G1 and G2 were retained in this study, because they have been determined as highly influential predictors in the feature selection process. Thus, the set of final features that were used to model consisted of the previous academic performance indicators and the chosen demographic and behavioral features.

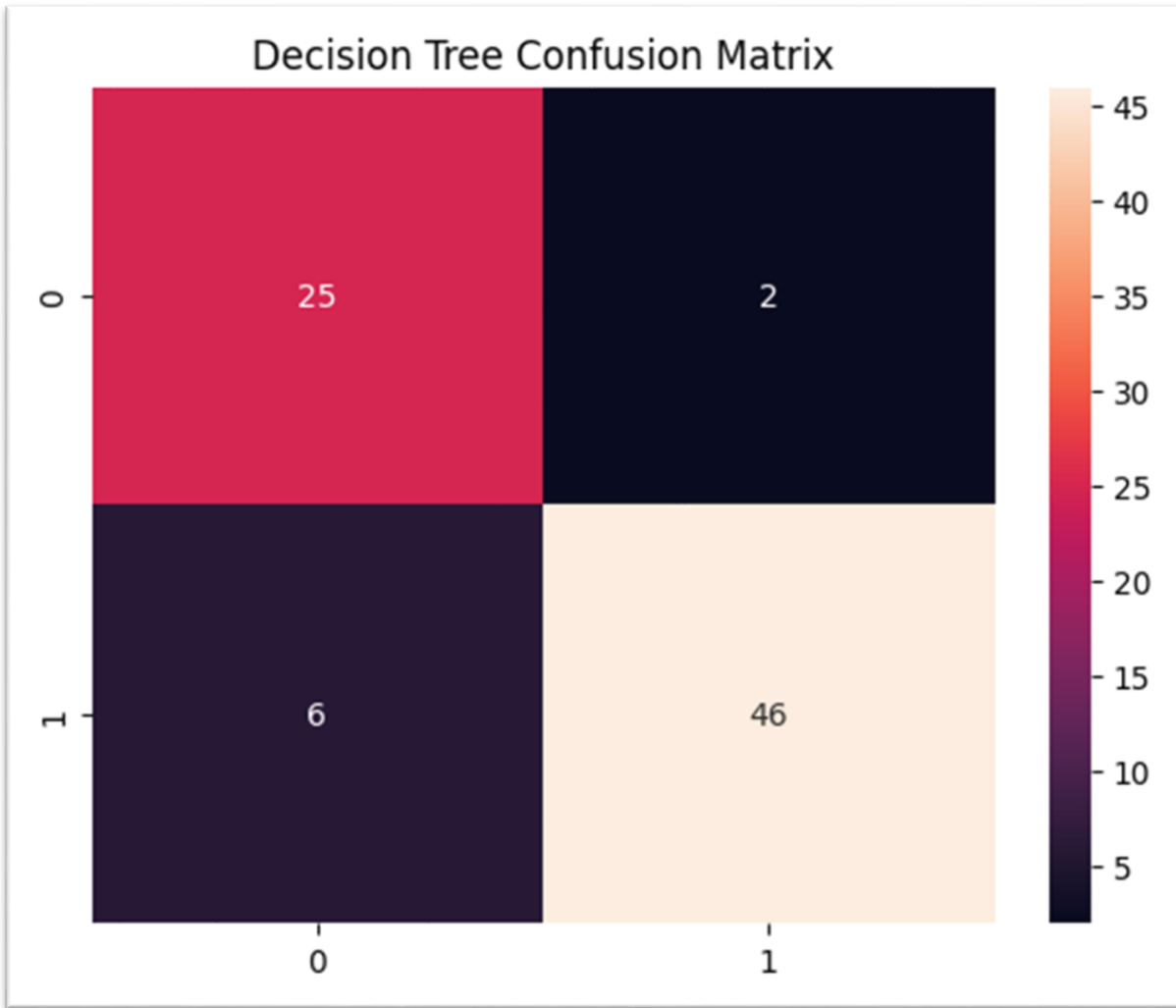
## 4.3 Model Performance

In this section, the performance of the individual machine learning models is provided in terms of various evaluation metrics.

### 4.3.1 Decision Tree

The accuracy of the Decision Tree model was around 0.8987, and this means that the model was able to classify nearly 90% of the cases correctly. The model was found to be very precise with the precision being 0.9583, which means that most of the predicted positive cases (students predicted to pass) were correct. The recall value of 0.8846, however, indicates that the model failed to detect some actual positive cases.

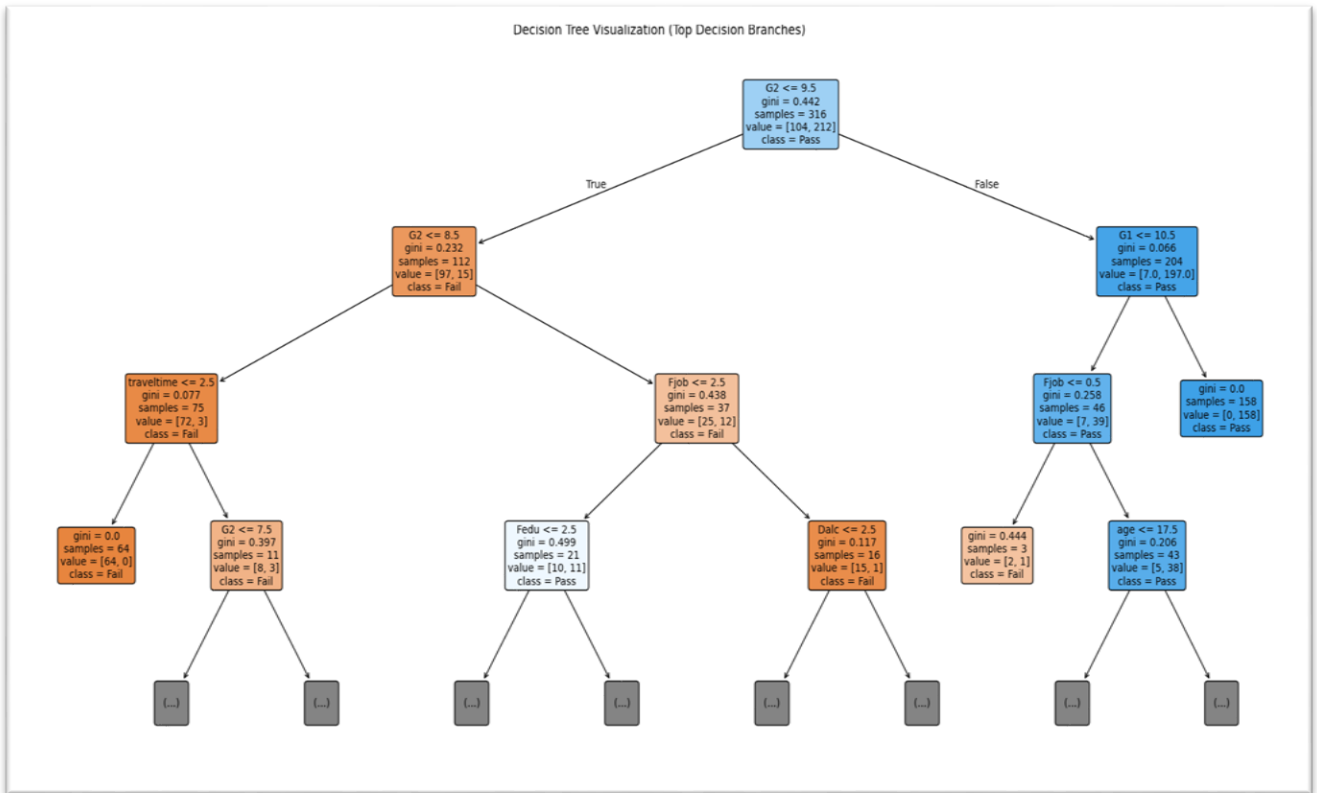
The F1-score of 0.92 indicates an excellent trade-off between precision and recall. This confusion matrix indicates that the model made accurate predictions in 46 cases on true positive and in 25 cases on true negative. It made errors in its predictions in 6 cases on false negative and in 2 cases on false positives. The AUC-ROC score of about 0.951 reflects a good level of discriminative ability.



*Figure 4.1 Decision Tree Confusion Matrix*

The confusion matrix shows that the Decision Tree model had a higher number of false negatives than false positives. This resulted in several students who truly were failing academically being labeled as passing students. Such mistakes can hinder the efficacy of early intervention in classroom environments, as certain students in need of academic help might not be addressed in a timely fashion. A relatively high precision value shows that if the student is predicted to pass, then the prediction is likely to be correct. However, recall value is low compared with other models, indicating that the model is less effective in identifying all at-risk students.

Moreover, the performance exhibited on the testing dataset is comparable to the F1 score computed during the model training phase from the cross-validation dataset. This means that the model is not showing significant signs of overfitting and has a reasonable ability to extrapolate to make predictions about data it hasn't seen before. However, the Decision Tree algorithm is slightly less stable than the other algorithms as it is sensitive to the changes in training data.



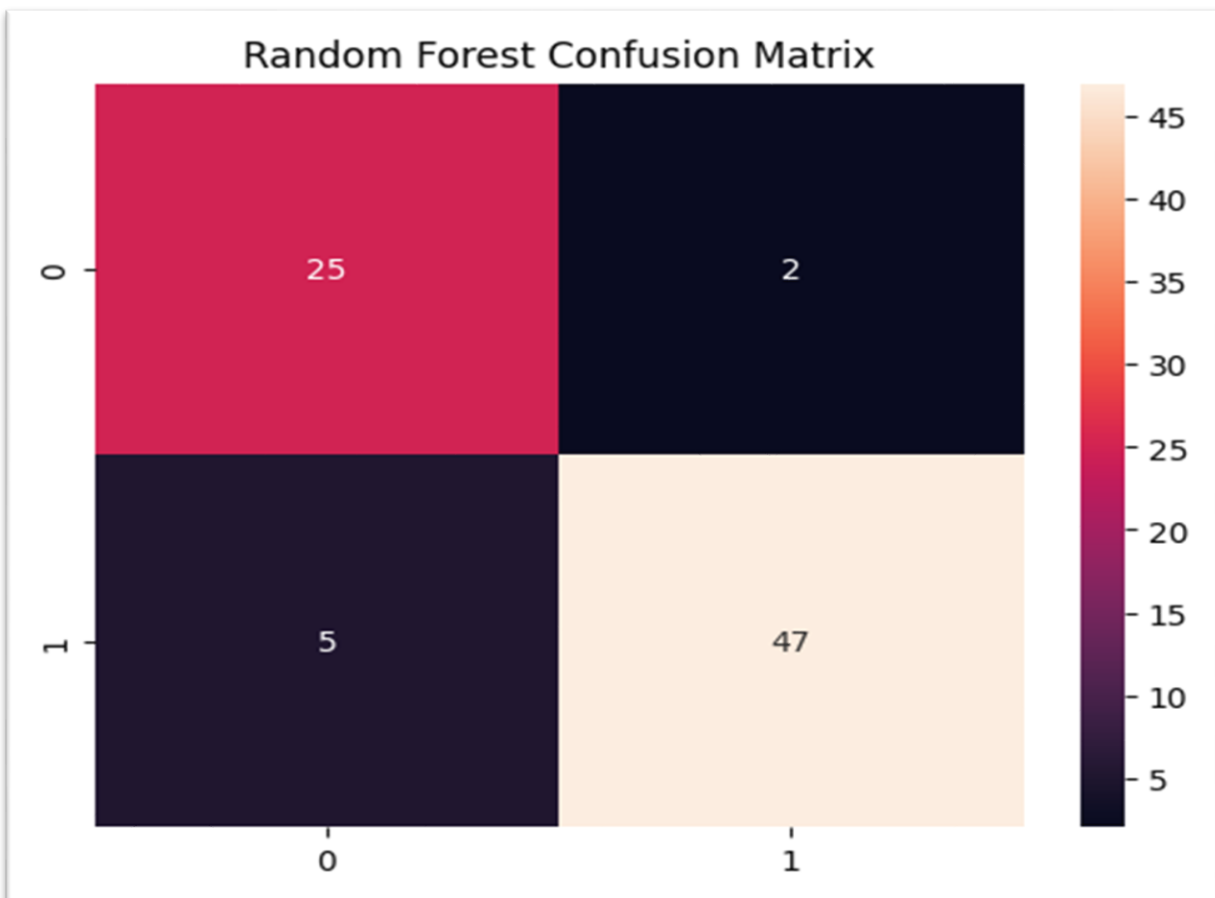
*Figure 4.2 Decision Tree Visualization*

The Decision Tree visualization provides an interpretable representation of how the classification model predicts student academic performance. The upper decision branches clearly demonstrate that G2 and G1 are the most influential predictive variables within the model. Students with higher previous academic grades were generally classified as successful students, while lower G2 values increased the likelihood of predicting academic risk. Secondary variables such as travel time, alcohol consumption (Dalc), parental education, and age also contributed to classification decisions at lower levels of the tree. The hierarchical structure of the model illustrates how the Decision Tree separates students into different outcome categories based on the most significant academic and behavioral predictors. This interpretability represents one of the major advantages of Decision Tree models because educators and academic administrators can visually understand how classification decisions are generated.

### 4.3.2 Random Forest

The Random Forest model was found to perform better than the Decision Tree model. It had a precision of 0.9592 and a recall of 0.9038 with an accuracy of about 0.9114. The F1-score of 0.9307 means that the balance and predictive performance are better.

As shown in the confusion matrix, the model has accurately classified 47 true positives and 25 true negatives, with fewer misclassifications than the Decision Tree model. The model is also highly effective in the distinction between classes as the AUC-ROC score of approximately 0.972 further confirms the high ability of the model to distinguish between classes.



*Figure 4.3 Random Forest Confusion*

The Random Forest model had a higher generalization and made fewer classification errors than the Decision Tree model. The confusion matrix shows that there were fewer false negatives which means that it was better at correctly identifying students that were at risk of poor academic performance. This is important in education prediction tasks as it reduces the risk of false

negatives, which would allow vulnerable students to fall through the cracks in institutional support systems.

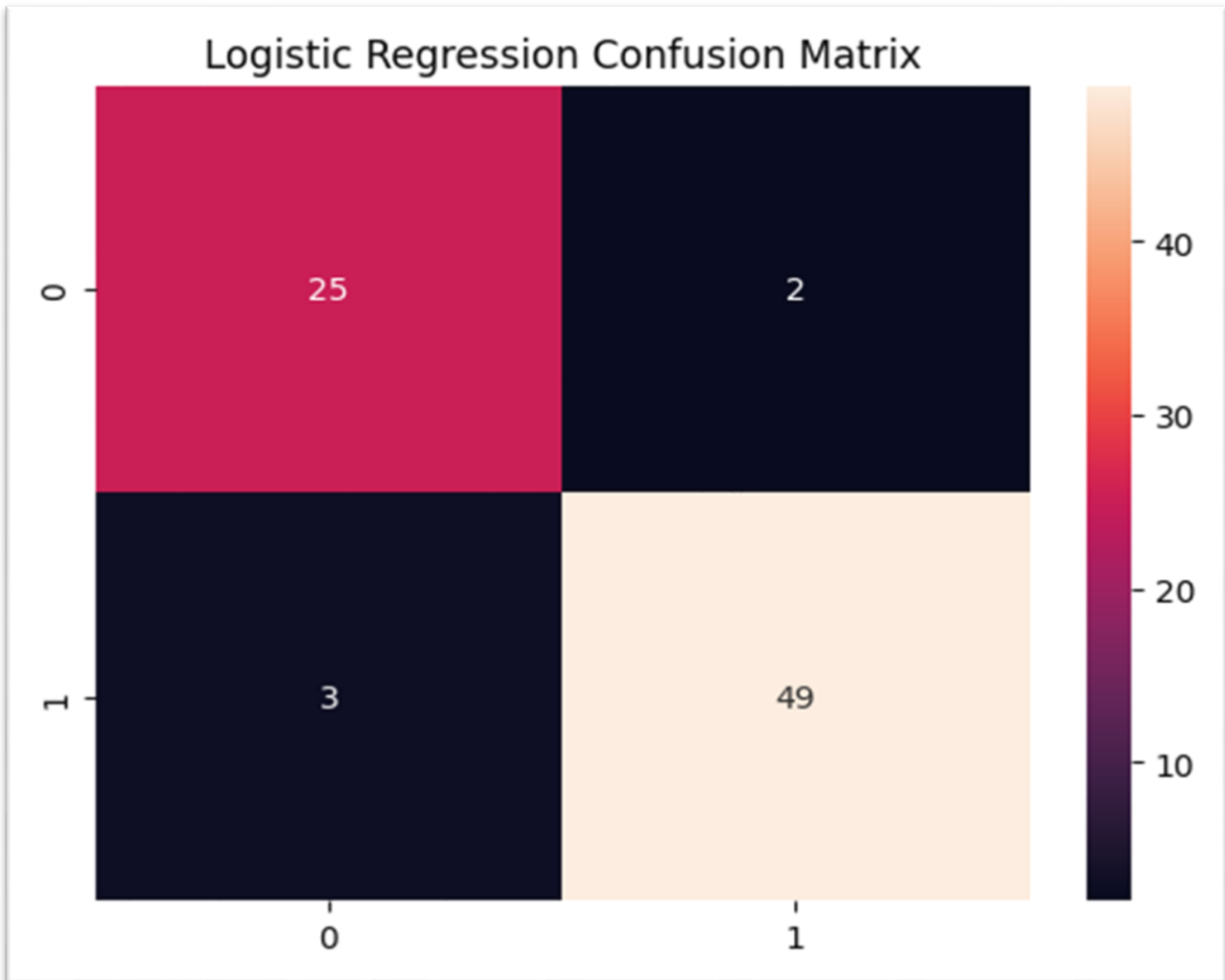
The performance improvement of the Random Forest could be explained by the ensemble learning approach, where several decision trees work together to make a prediction. This technique helps to minimize model variance and enhance the stability of the prediction made by the model when compared to a single Decision Tree model. Furthermore, the relatively good precision and recall scores indicate that the model is effective at both positive and negative classification. The similarity between the results of the cross-validation and the testing performance also points in this direction, suggesting that cross validation showed a very good generalization, and it was concluded that there was no significant overfitting of the model.

### 4.3.3 Logistic Regression

The best result of all other models was the Logistic Regression. The Logistic Regression model resulted in the highest accuracy with a value of around 0.9367. It also showed the best precision (0.9608) and recall (0.9423) with an F1 score value (0.9515).

The confusion matrix indicates that Logistic Regression was able to classify 49 true positives and 25 true negatives correctly with only 3 false negatives and 2 false positives. The AUC-ROC of about 0.981 indicates that this model is best of the three in that it has an excellent classification capability.





*Figure 4.4 Logistic Regression Confusion Matrix*

Logistic Regression performed best overall, especially in terms of recall and AUC-ROC, out of the three machine learning models. The confusion matrix reveals that the model made the least number of false negatives, indicating that the model is very effective at identifying students of true risk of poor academic performance. This is very useful in the practical education setting, as it lowers the number of false negatives and enables timely academic intervention for students who need it.

Also, Logistic Regression outperforms other classifiers, possibly because of the high linear correlation between G2 and G3, therefore the classification problem is almost linearly separable. This made Logistic Regression an efficient modeler without adding complexity to the Tree-based models. In addition, there is a good degree of consistency between the performance of the Cross Validation and Testing sets, which indicates that the model for Logistic Regression did not show significant overfitting.

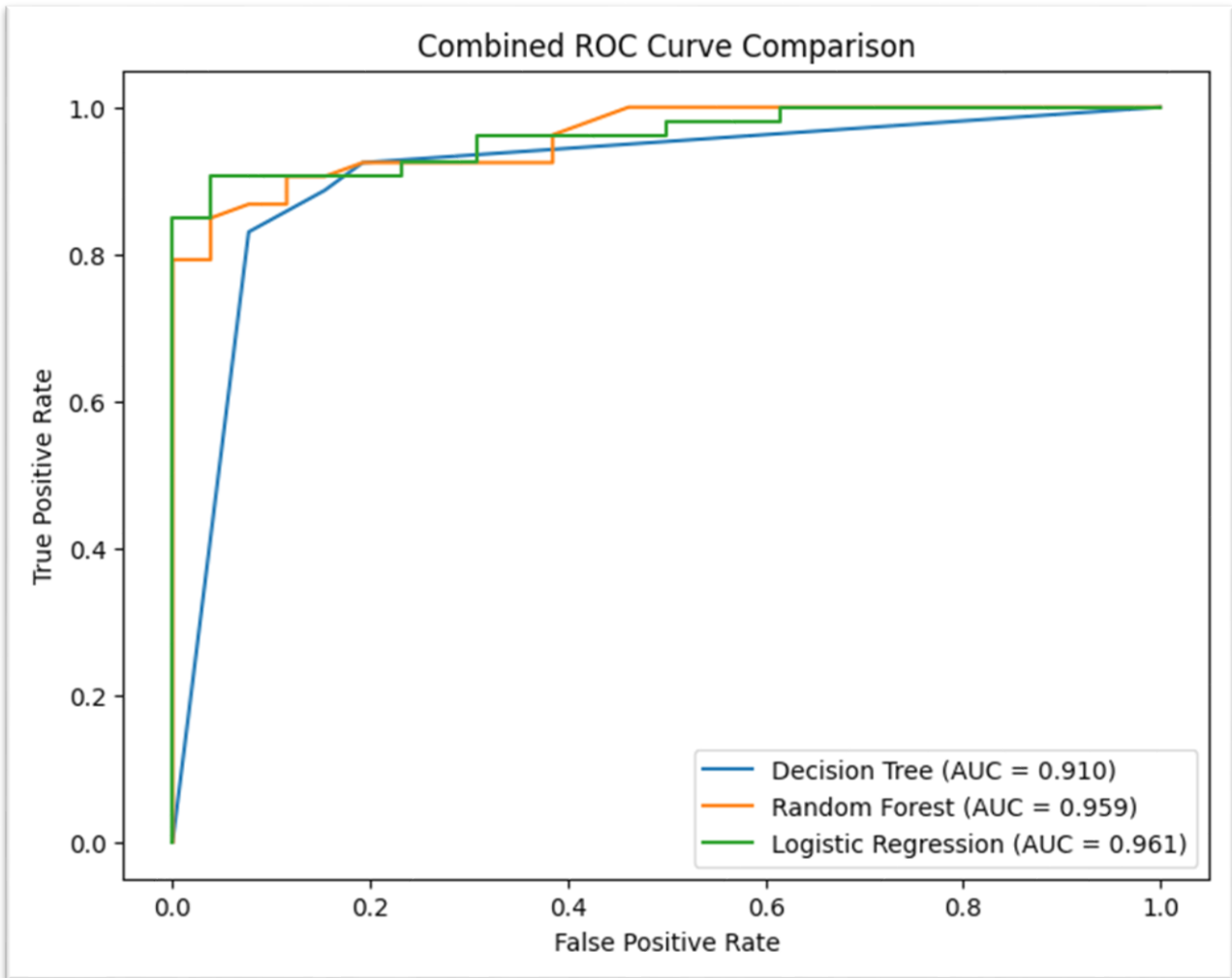
## 4.4 Cross-Model Analysis

The three models were compared and contrasted to determine the relative performance of the three models.

### Comparative Model Performance Table

| Model               | Accuracy | Precision | Recall | F1-Score | AUC_ROC |
|---------------------|----------|-----------|--------|----------|---------|
| Logistic Regression | 0.9367   | 0.9608    | 0.9423 | 0.9515   | 0.9808  |
| Random Forest       | 0.9114   | 0.9592    | 0.9038 | 0.9307   | 0.9722  |
| Decision Tree       | 0.8987   | 0.9583    | 0.8846 | 0.9200   | 0.9512  |

The findings clearly show that Logistic Regression is superior to the others in all measures of evaluation. Random Forest will be number two, and the Decision Tree will be rather lower in performance, still acceptable.



*Figure 4.5 Combined ROC Curve Comparison*

These results are also expressed by the “combined ROC curve” in Figure 4.4. From the ROC comparison, it is seen that the Logistic Regression model has the highest curve among the three machine learning models, Decision Tree and Random Forest. This means that Logistic Regression has the highest overall classification power and the highest classification power at all classification thresholds between the pass and fail classes.

The Random Forest model also had good discriminatory capabilities, but inferior to Logistic Regression. On the other hand, the ROC curve of the Decision Tree model is the lowest of the three models, which shows that it has a relatively poor classification ability. In general, Logistic Regression has the best predictive accuracy and stability in predicting student academic performance, as revealed by the combined ROC analysis.

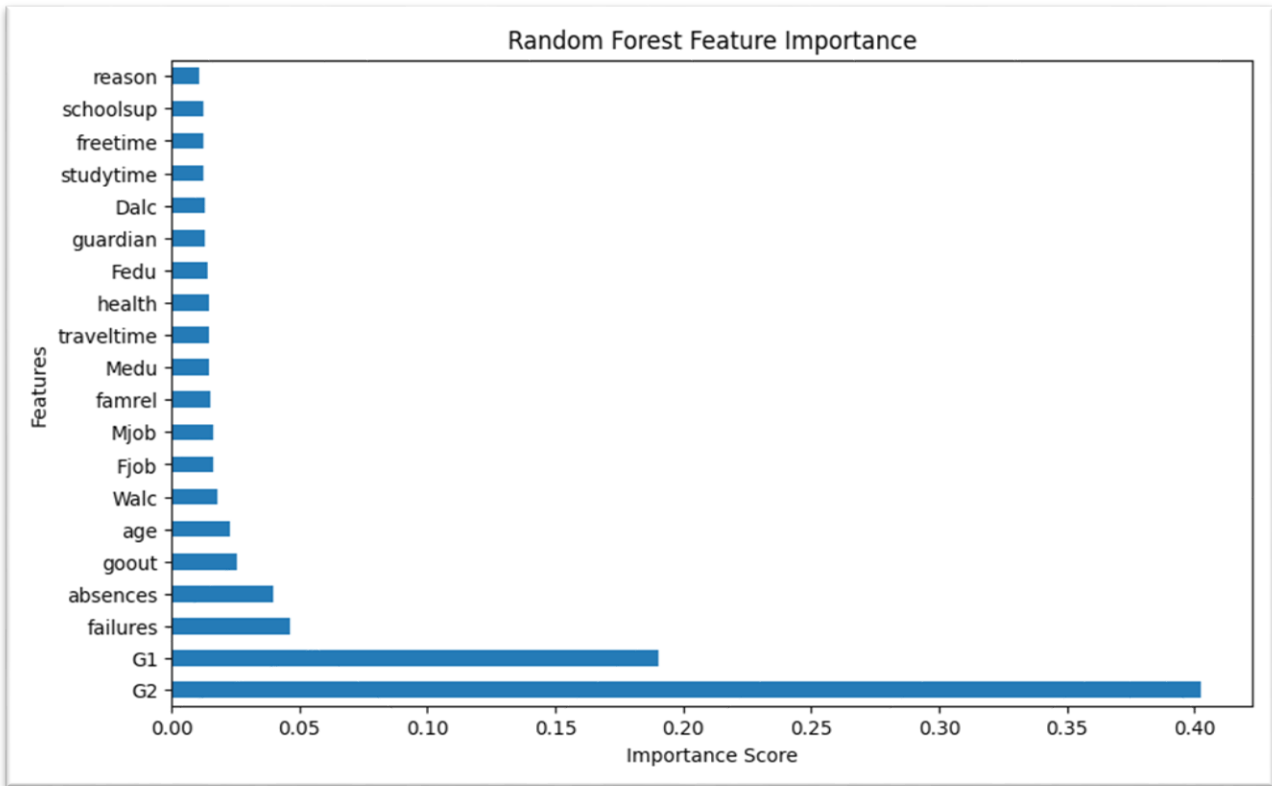
## 4.5 Feature Importance Analysis

The analysis of the most significant predictors of student performance was done using the Random Forest model. The best features that were identified are G2, G1, absences, age, and failures.

The dominances of G2 and G1 are very much related to educational theories presented in Chapter 2. The constructivist learning theory suggests that prior knowledge is a very important factor to consider when determining future learning outcomes. This point of view is supported by the strong influence of the previous grades (G1 and G2) which demonstrates that academic performance is cumulative.

The variable “absences” is a measure of student engagement and attendance which can be attributed to behavioral theories like self-efficacy. When the students attend classes regularly, there are high chances that they would build confidence and achieve better academic performance. Equally, the variable “failures” represents a history of poor performance in the past, and this may adversely affect future performance.

The fact that age is a very significant aspect implies that maturity and learning levels could also have an impact on academic performance. In general, the obtained results of feature importance offer valuable information on the variables that contribute to student performance and are consistent with the theoretical expectations.



*Figure 4.6 Random Forest Feature Importance*

In addition to the highly influential variables, several features such as internet access, romantic relationship status, extracurricular activities, and family support demonstrated relatively low importance scores in the Random Forest model. Although these variables are commonly assumed to influence student learning outcomes, their predictive contribution within this dataset was comparatively limited. One possible explanation is that academic performance in this dataset is more strongly influenced by direct academic indicators such as previous grades and failure history than by broader social or lifestyle variables.

The findings of this study are generally consistent with the work of Paulo Cortez and Alice Silva, who also identified previous academic grades as the strongest predictors of student performance. Similar to the present study, their research demonstrated that G1 and G2 possess substantial predictive influence on final academic outcomes. This consistency strengthens the reliability of the current findings and supports the theoretical assumption that student academic performance is cumulative in nature.

However, an important methodological limitation associated with Random Forest to feature importance should also be acknowledged. Gini-based feature importance measures may exhibit bias toward numerical variables or variables with higher cardinality. The result is that some variables like G1 and G2 can be given disproportionately high importance scores due to the

greater level of numerical variability. So, the importance of features should be taken a bit more with a pinch of salt and not as definitive indicators of what causes the behavior.

In general, the feature importance analysis gives insight into the most important factors affecting the prediction of students' academic performance. The findings also confirm the theoretical underpinnings presented in Chapter 2, including the role of previous academic performance, student involvement, and behavioral tendencies in explaining subsequent academic performance.

## 4.6 Summary

This chapter displayed the findings and discussion of the machine learning models that were used to predict academic performance of students. The outcomes of the preprocessing made certain that the data was well-prepared and balanced between the training and testing sets. Logistic Regression was the best overall model of the models that were evaluated, followed by random forest and decision tree.

The results indicate that previous academic achievement and behavioral variables play a crucial role in predicting student outcomes. The results also demonstrate the effectiveness of machine learning techniques in educational data analysis. In the following chapter, conclusions and recommendations will be drawn out of these conclusions.

# Chapter 5 – Conclusions, Recommendations & Future Directions

## 5.1 Conclusion

The objective of this study was to identify the academic performance of students by using machine learning methods from demographic, academic, and behavioral factors. The Data source for the research was the Student Performance Data from Kaggle which includes students' educational and social background, study habits, and prior academic performance. Academic grades (G3) were selected as the targeted variable, and it was converted to a binary classification problem representing pass or fail outcomes.

The research aims were achieved by using three machine learning algorithms: Decision Tree, Random Forest, and Logistic Regression. Before model development, data was processed by several preprocessing techniques like data cleaning, label encoding, feature selection, feature reduction, and train test splitting by stratified sampling. Several feature selection methods were

used to determine the most influential features affecting students' performance, including correlation analysis, Chi-squared testing, and the use of Random Forest feature importance.

From the study, it can be concluded that academic performance variables are the best predictors of the final student outcome with prior academic performance (G1 and G2) being the strongest. Absences, failures, and age were also important predictors of accuracy. The results align with previous learning theories mentioned in Chapter (2), such as constructivist learning theory and self-efficacy theory, both of which focus on the role of learning experiences and students' involvement in learning outcomes.

In logistic Regression, they have also shown the highest accuracy, precision, recall, F1 score, and AUC-ROC value among the three machine learning models tested, making it the best overall model. Random Forest ranked second and Decision Tree third among the three machine learning models, although both models still achieved acceptable predictive performance levels. The results show the potential of machine learning methods for predicting student academic achievement and offering valuable insights for educational institutions.

This is one of the key results that this study has uncovered: Logistic Regression achieves better performance than the tree-based models on all key evaluation metrics, such as Accuracy, Recall, F1-score, and AUC-ROC. This result indicates that most of the predictive relations in the data are linear, especially between G2 and the final academic grade (G3), since there is a very high association between these two. As a result, Logistic Regression is better able to capture the relationship between features than other more complicated tree-based algorithms, and the classification problem is closer to being linearly separable. This result is also broadly in line with earlier studies on educational data mining, for example Cortez and Silva (2008) who discovered that students' previous academic performance was the most reliable predictor of later academic performance. Thus, findings of the present study confirm the significance of previous academic achievement measures in predicting educational outcomes.

Finally, the research aimed to meet the goals and objectives and showed the effectiveness of machine learning techniques in predicting student academic performance. Results showed that G1 and G2 were the most important academic achievement variables to make predictions for final academic results. The machine learning algorithms evaluated were Logistic Regression, which gave the highest overall predictive performance, indicating that the predominant predictive relationships in the data are mostly linear. The results of this study support previous research in the field of educational data mining, which has focused on the value of previous academic achievement as a predictor of future educational success. The study clearly highlights the utility of these predictive analytics and machine learning methods to academic monitoring, early identification of at-risk students, and evidence-based academic decision-making.

## 5.2 Recommendation

This study concludes with some recommendations which can be followed by education institutes, researchers, and policy makers.

In the first place, there is a need to investigate implementing predictive analytics systems in educational institutions to observe student academic performance from an early stage. Identifying previous grades and attendance as good predictors of success provides institutions with a tool for identifying students who might need extra academic assistance or intervention.

In the second place, the education providers should aim to increase students' engagement and attendance. Besides, the school attendance was shown to adversely affect learning outcomes. Academic advising, mentorship, and individual learning plans can help enhance student engagement and learning outcomes.

Thirdly, it is important to encourage the use of data-driven decision making in the management of academic institutions. Machine learning models will help the administrators and educators to understand the learning process and behavior of students and take sound decisions on education matters.

Furthermore, other researchers conducting similar studies should use larger and diverse educational datasets to increase the generalizability of the model. A wider view of student performance prediction can be achieved by having students from various levels, institutions, and geographical areas.

Lastly, the results of this study show that previous academic performance variables, especially G1 and G2, were the most significant variables to make prediction for final student outcomes. Hence, it is essential for educational institutions planning to adopt predictive analytics systems to carefully gather, normalize, and track interim academic achievement data, which contributes most to predictive accuracy. Institutional Data Management practices to enhance continuous academic assessment can have a significant impact on the effectiveness of student performance prediction and early intervention systems.



## 5.3 Future Directions

This study yielded encouraging outcomes, but there were opportunities for further research and improvements.

For future studies, other machine learning and deep learning techniques can be employed, including Support Vector Machine (SVM), Gradient Boosting, XGBoost, Artificial Neural Network (ANN), and Deep Neural Network (DNN). This would make predictions much more precise and capable of handling intricate educational data.

Furthermore, it would be beneficial to use actual learning data in real time education data from learning management systems (LMS), online learning platforms, and classroom activities in future studies. Real-time behavioral data can enhance the prediction potential and facilitate ongoing academic monitoring.

Another potential direction is the creation of a real Early Warning System (EWS) that in addition to predicting student performance, it also suggests targeted interventions. These systems could give automatic alerts to teachers and academic advisors when they have identified a student as being “at risk.”

Further research could also explore other variables that impact student performance, such as psychological variables, motivation, mental health, socioeconomic factors, and learning styles. These factors may provide an understanding of academic achievement more fully.

Moreover, further studies might explore whether techniques of explainable artificial intelligence (XAI) could be applied to improve model interpretability and transparency. Explainable models can better inform educators why certain predictions are made and increase trust in machine learning systems.

Finally, there are cross-institutional and longitudinal studies that may provide stronger evidence of the long-term effectiveness of predictive models in education. This study could help build more powerful and scalable student performance prediction systems.

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